

## Data Flow Diagram

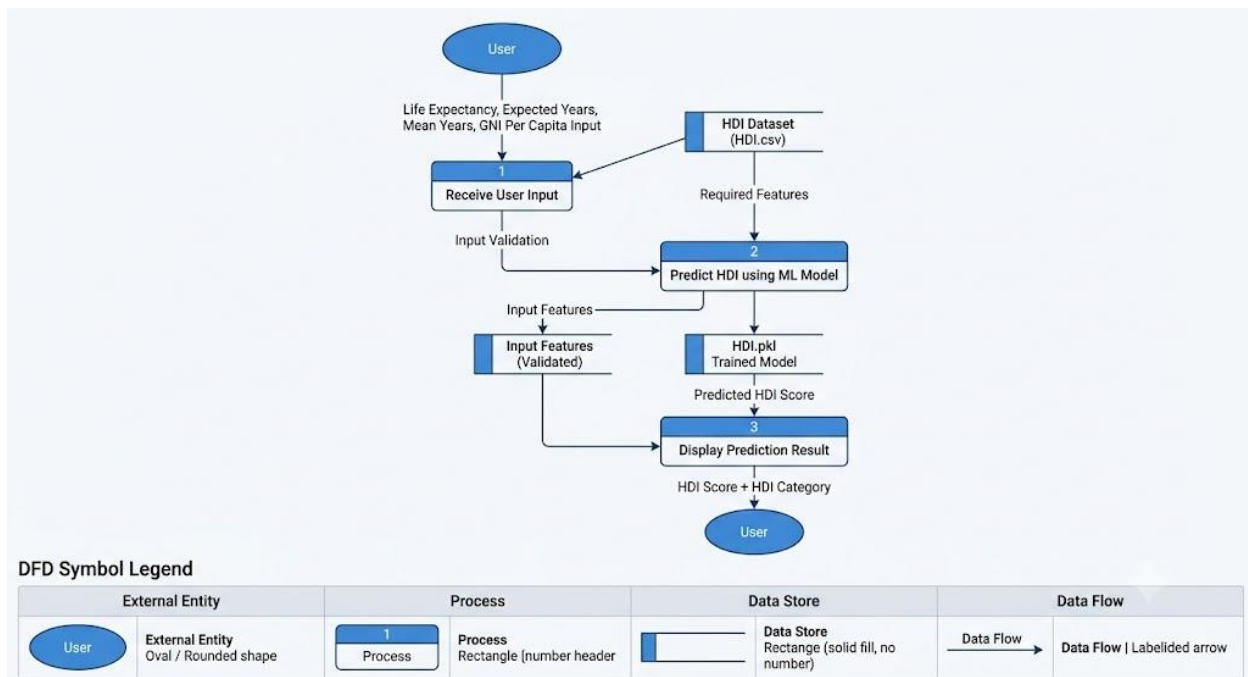
|                      |                                              |
|----------------------|----------------------------------------------|
| <b>Date</b>          | 15 March 2026                                |
| <b>Team ID</b>       |                                              |
| <b>Project Name</b>  | <b>A Comprehensive Measure of Well-Being</b> |
| <b>Maximum Marks</b> | 2 Marks                                      |

### Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

#### DFD Symbol Legend

| Symbol                                   | Name                   | Description                                                                                                          |
|------------------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Oval / Rounded shape</b>              | <b>External Entity</b> | A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway). |
| <b>Rectangle with numbered header</b>    | <b>Process</b>         | An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).                |
| <b>Rectangle (solid fill, no number)</b> | <b>Data Store</b>      | A place where data is held for later use (e.g., Customer Database, Inventory).                                       |
| <b>Labeled arrow</b>                     | <b>Data Flow</b>       | The movement of data between entities, processes, and data stores, labeled with what data is being passed.           |



## ➤ DATA FLOW DIAGRAM (DFD)

- **External Entity**
  - User

### ➤ Processes

#### Process 1: Receive User Input

The user enters the required Human Development Index (HDI) parameters through the web interface (index.html).

#### Input Parameters:

- Life Expectancy at Birth
- Expected Years of Education
- Mean Years of Education
- Gross National Income (GNI) per Capita

#### Process 2: Predict HDI using Machine Learning Model

The Flask application (app.py) validates the user's input, loads the trained machine learning model (HDI.pkl), and predicts the Human Development Index (HDI) score.

#### Process 3: Display Prediction Result

The predicted HDI score and its corresponding HDI category (Very High, High, Medium, or Low) are displayed to the user through the result.html page.

### Data Stores

#### D1 – HDI.csv

Stores the Human Development Index dataset used for training the machine learning model.

**D2 – HDI.pkl**

Stores the trained Linear Regression model used to predict the Human Development Index.

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**Data Flows**

| From      | To        | Data Flow                            |
|-----------|-----------|--------------------------------------|
| User      | Process 1 | HDI Input Parameters                 |
| Process 1 | Process 2 | Validated Input                      |
| HDI.csv   | Process 2 | Training Data                        |
| HDI.pkl   | Process 2 | Trained Machine Learning Model       |
| Process 2 | Process 3 | Predicted HDI Score                  |
| Process 3 | User      | Predicted HDI Score and HDI Category |

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## Project Flow

